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Using Davis's Perceived Usefulness and Ease-of-use Instruments for Decision Making: A Confirmatory and Multigroup Invariance Analysis

William J. Doll
Department of Management, College of Business, The University of Toledo, 2801 West Bancroft Street, Toledo, OH 43606, e-mail: william.doll@utoledo.edu

Anthony Hendrickson
Department of Management, College of Business, Iowa State University, Ames, IA 50011-2063

Xiaodong Deng
Department of Information Systems and Operations Management, College of Business, The University of Toledo, 2801 West Bancroft Street, Toledo, OH 43606

ABSTRACT

As key components of Davis's technology acceptance model (TAM), the perceived usefulness and perceived ease-of-use instruments are widely accepted among the MIS research community as tools for evaluating information system applications and predicting usage. Despite this wide acceptance, a series of incremental cross-validation studies have produced conflicting and equivocal results that do not provide guidance for researchers or practitioners who might use the TAM for decision making. Using a sample of 902 "initial exposure" responses, this research conducts: (1) a confirmatory factor analysis to assess the validity and reliability of the original instruments proposed by Davis, and (2) a multigroup invariance analysis to assess the equivalence of these instruments across subgroups based on type of application, experience with computing, and gender.

In contrast to the mixed results of prior cross-validation efforts, the results of this confirmatory study provide strong support for the validity and reliability of Davis's six-item perceived usefulness and six-item ease-of-use instruments. The multigroup invariance analysis suggests the usefulness and ease-of-use instruments have invariant true scores across most, but not all, subgroups. With notable exemptions for word processing applications and users with no prior computing experience, this research provides evidence that the item-factor loadings (true scores) are invariant across spread sheet, database, and graphic applications. The implications of the results for managerial decision making are discussed.

Subject Areas: Confirmatory Factor Analysis, Factorial Invariance, Instrument Validation, Perceived Ease of Use, Perceived Usefulness, Research Methods, and Technology Acceptance Model.

INTRODUCTION

Organizational investments in information technology to support planning, decision making, and communication processes are inherently risky. To make effective use of this technology, managers, professionals, and operating employees must accept the applications, learn how to interact directly with aspects of the hardware and software, and adapt the technology to their tasks requirements. Understanding why people accept or reject computers has proven to be one of the most challenging issues in information systems research (Swanson, 1988). Davis (1986) introduced the technology acceptance model (TAM) to explain and predict computer usage behavior. TAM can be used to evaluate systems very early in their development, or to assess user reactions to systems on a trial basis in advance of purchase decisions (Davis, Bagozzi, & Warsaw, 1989).

TAM posits that two particular beliefs, perceived usefulness and perceived ease of use, can predict computer acceptance behavior. Perceived usefulness is defined as the prospective user's subjective probability that using a specific application system will increase his or her job performance within an organizational context. Perceived ease of use refers to the degree to which the prospective user expects the target system to be free of effort.

Perceived usefulness and ease of use are meant to be fairly general determinants of user acceptance. Davis et al. (1989) described them as belief sets that are meant to be readily generalizable to different computer systems (applications) and user populations. They argued that the ability to take robust, well-formed measures of the determinants of user acceptance early in the development process (i.e., initial exposure data) can have an impact on system acceptance by enabling developers to weed out bad systems, refine the rest, and generally cut the risk of delivering finished systems that get rejected by users.

Davis provided strong theoretical support for his perceived usefulness and ease-of-use instrument(s) and used a rigorous development methodology (Davis et al., 1989; Davis, 1989). The instruments are widely regarded and have received considerable attention by researchers (Adams, Nelson, & Todd, 1992; Chin & Todd, 1995; Hendrickson, Massey, & Cronan, 1993; Igbaria, Guimaraes, & Davis, 1995; Mathieson, 1991; Segars & Grover, 1993; Straub, Limayem, & Karahanna-Evaristo, 1995; Subramanian, 1994). Davis (1989), Adams et al. (1992), and Subramanian (1994) have suggested that these instruments might be used broadly within and across organizations to evaluate applications (technologies), make comparisons between user groups or applications, and predict usage. To make such comparisons, the instruments must be valid and reliable; they must also be robust across applications (technologies) and demographic categories of users.

If an instrument is robust, its value for practical decision making and research is greatly enhanced (Bejar, 1980; Gorsuch, 1970). An instrument can be considered robust if its items have equivalent meaning (i.e., equal item-factor loadings or true scores) across subgroups or conditions of measurement. Standardized instruments must provide equivalent (invariant) measurement if comparative statements are to have substantive import. Without equivalent measurement, observed scores from different groups are in different scales and, therefore, are not directly comparable (Drasgow & Kanfer, 1985). These ideas are particularly clear

in physical measurement, where it is obvious that length in inches cannot be directly compared to length in centimeters. Likewise, in instruments used in MIS research, observed scores cannot be compared unless they are in the same scale.

Many instruments, especially those that measure individual differences, are group or context specific in the way they operate. In a review and critique of the MIS instruments, Klenke (1992) argued that where data are collected with reference to a specific application, an instrument's measurement equivalence (invariance) across applications or subpopulations is an important issue. Thus, she called for multigroup invariance studies of instruments such as perceived usefulness and ease of use, in which data are gathered with respect to a particular application or technology.

Multigroup analysis of factorial invariance can be used to test the measurement equivalence of an instrument. Confirmatory factor analysis permits more rigorous tests of the equality or invariance of measurement parameters (e.g., item-factor loadings) across groups than are possible with exploratory factor analysis (Alwin & Jackson, 1981; Jöreskog & Sörbom, 1993; Joreskog, 1979; Marsh & Hocevar, 1985). Testing for invariance (i.e., equivalence) is a particularly demanding test of an instrument's robustness. For some purposes, minor differences in item-factor loadings may not be critical to the interpretation of research results (Byrne & Shavelson, 1987).

Davis's (1989) perceived usefulness and ease-of-use instruments have the promise of being broadly applicable across technologies and organizations, but their measurement equivalence has not been tested. Thus, we know little about the robustness (i.e., measurement equivalence) of these instruments. Using a sample of 902 respondents, this study first conducts a confirmatory factor analysis to assess the validity and reliability of Davis's proposed instruments. Next, multigroup analysis of factorial invariance is used to assess the measurement equivalence of the perceived usefulness and ease-of-use instruments across type of application, level of experience, and gender.

THE SUBSTANTIVE RESEARCH QUESTION

Constructs serve as the language through which theoretical ideas and research findings are communicated among researchers and practitioners. The usefulness of a construct is determined by the importance of its domain, the breadth of its applicability, and the value added by the construct. Constructs clearly should be relevant to a variety of domains within a field of study or of critical importance to a single domain. If a construct's relevance is limited, it will not attract the attention of many researchers and practitioners in the field and, hence, will not inform the future direction of research and practice in its domain. Many, if not most, of the measurement scales developed by IS researchers have been designed to be general (i.e., applicable across a wide variety of situational contexts). Designing instruments for broad applicability increases the extent to which the results of studies can be generalized to other contexts. It also facilitates the ability to interpret results as informing the full body of theory to which the studies are directed (Zmud, Sampson, Reardon, Lenz, & Byrd, 1994).

The establishment of construct validity within a population does not assure the equivalence of measures and constructs across subgroups. When practitioners or researchers make comparisons between the mean scores for groups of respondents, the implicit assumption is that the items that scale the construct have the same meaning in each group. When data from different groups of respondents are combined to study the relationship between various causal factors and some dependent variable, an underlying assumption is that all variables have the same meaning across subgroups.

The practical application of an instrument may depend upon its robustness. Time constraints and parsimony attract managers to utilize a single measurement scale to assess users' perceptions of usefulness and ease of use across heterogeneous software applications. If an instrument is not robust, the comparison of scores between groups of respondents is difficult, if not impossible. Using an instrument whose scale differs between subgroups can result in poor managerial decisions or incorrect statistical inferences.

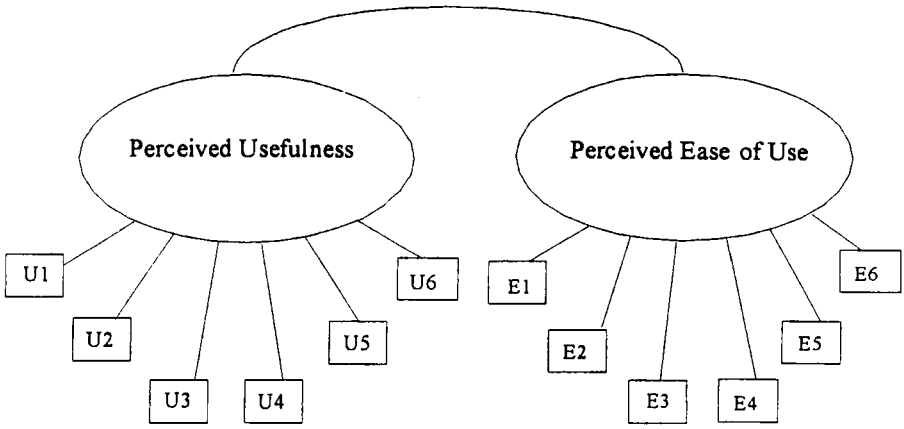
Zmud et al. (1994) argued that all instruments could be viewed as located on a continuum reflecting the extent to which the construct is linked with an experientially based context. The stronger a construct's linkage to an experientially based context, the greater the concern that the construct and the context may interact. If significant construct-context interaction is present, item-factor loadings or uniqueness (error) can be expected to vary between subgroups or contexts. Thus, the context may, in part, influence the construct's meaning or how it is scaled.

Smith, Kendall, and Hulin (1969) suggested that measurement nonequivalence might result from the use of different frames of reference by individuals in different situations. The frame of reference an individual brings to bear on evaluating an application is shaped by many factors, including role expectations for their behavior, their task requirements, or individual skills and beliefs. These differing frames of reference are based upon the unique life experiences of demographic categories of respondents. They can also be attributed to experientially based differences in the task/work context of respondents across types of applications or conditions of measurement.

Previous Cross-validation Studies

Previous studies using these instruments have questioned the model-data fit of Davis's original measurement model (see Figure 1) and the construct validity of individual items. Table 1 profiles these studies in terms of sample size, type of data, application type, and the observations and/or issues raised by the study. Table 2 describes the items recommended/used in each study. Cross-validation efforts have sampled users of existing applications rather than gather data from individuals who report their beliefs/opinions after a brief, and possibly their first, exposure to the application. People who have prior application-specific experience might be expected to have well-formed beliefs about the application. Whether users can form such beliefs on the basis of only a brief exposure is more problematic. Brief exposure data are important to assessing whether the technology acceptance model can be used to make early evaluations in the development process, or to assess user reactions on a trial basis in advance of purchase decisions.

Figure 1: Perceived usefulness and ease-of-use measurement model.



Cross-validation efforts have focused primarily on "communication" applications such as email, vmail, and dial-up. Limited cross-validation evidence is available for widely used "office suite" applications such as database, spreadsheet, presentation graphics, and word processing. Previous studies have focused on a single application or analyzed two samples separately rather than combining observations across technologies or user populations. This practice of analyzing each type of application separately may reflect an underlying concern that the scores may not be comparable across different applications or user characteristics. Some of the studies reported gender and application-specific experience, but they did not analyze the data by these categories. None of the studies reported the respondents' prior general computing experience (i.e., general experience with computing rather than experience with the specific application in question).

A pattern of incremental cross-validation studies (see Table 2) has produced conflicting and equivocal results with each study recommending that different items be used/eliminated (Adams et al., 1992; Chin & Todd, 1995; Segars & Grover, 1993; Subramanian, 1994). Among the six usefulness items, only two (U3 and U5) were recommended/used across all studies. Among the six ease-of-use items, only two (E1 and E6) were recommended/used across all cross-validation studies. Each of these studies attributed the measurement problems to different items, suggesting that results may be sample specific. The results for the structural model have also been equivocal (Chin & Todd).

With the notable exception of Chin and Todd (1995), most of these cross-validation studies used samples considered small for confirmatory factor analysis. Harris and Schaubroeck (1990) suggested a minimum sample size of 100 for confirmatory analysis, but recommended a sample size of at least 200. Sample sizes of 67 or less were used to cross-validate graphic, word processing, and spreadsheet applications that are widely used in business.

Some of these cross-validation studies modified Davis's original measurement model using a data-driven strategy rather than hypothesizing and testing alternative a priori measurement models. The issue of capitalization on chance must be

Table 1: The summary of previous cross-validation studies.

Studies	Subjects	Sample Size	Initial Exposure Data?	Application	Prior Application-Specific Experience	Gender	Observations/Issues/Findings
Adams et al., 1992 (Study 1)	Respondents from 10 business organizations	116	No	Email	Experienced (average 21 months)	41% female 59% male	1. Sample was separately analyzed by type of application 2. Gender was reported but not analyzed separately 3. Eliminated two items for measuring ease of use 4. Revised model had marginal model-data fit.
		68	No	Vmail	Experienced (average 28 months)	40% female 60% male	
Adams et al., 1992 (Study 2)	Business students	64	No	Word Processing	Experienced (average 28 months)	(Not reported)	1. Sample was separately analyzed by type of application 2. Replaced one item in ease-of-use instrument 3. Revised model had unsatisfactory model-data fit 4. Found inconsistency between applications in the relative importance of usefulness & ease of use.
		67	No	Spreadsheet	Experienced (average 33 months)		
		54	No	Graphics	Experienced (average 13 months)		

Table 1: (continued) The summary of previous cross-validation studies.

Studies	Subjects	Sample Size	Initial Exposure Data?	Application	Prior Application-Specific Experience	Gender	Observations/Issues/Findings
Segars and Grover, 1993*	Respondents from 10 business organizations	116	No	Email	Experienced (average 21 months)	41% female 59% male	1. Sample was separately analyzed by type of application 2 Gender was reported but not analyzed separately 3. Two-factor model had marginal model-data fit 4. Recommended three-factor model with eight items had good model-data fit 5. Instability of structural model across applications may indicate underlying measurement problems 6. Measurement model may differ by application type or user characteristics.
		68	No	Vmail	Experienced (average 28 months)	40% female 60% male	
Subramanian, 1994	Respondents from two business organizations	75	No	Vmail	(Not reported)	(Not reported)	1. Sample was separately analyzed by type of applications 2. Two-factor model with seven items had a better model-data fit than the three-factor model 3. Two-factor measurement model is robust across two "communication" applications.
		104	No	Dial-Up			
Chin and Todd, 1995	Respondents from single business organization	259	No	Vmail	Novice (average less than one month)	(Not reported)	1. Found no empirical support or substantive rationale for separation of usefulness into two constructs 2. Recommended five items for measuring usefulness 3. Suggested caution in data-driven analysis 4. Emphasized need for substantive knowledge to drive model modification.

*Used the same data set in Adams et al.'s (1992) Study 1.

Table 2: Description of items recommended/used.

Item	Item Description	Items Recommended/Used				
		Adams et al. (Study 1, 1992)	Adams et al. (Study 2, 1992)*	Segars and Grover (1993)**	Subramanian (1994)	Chin and Todd (1995)***
Perceived Usefulness						
U1	Using <application name> in my job would enable me to accomplish tasks more quickly.	✔	✔			
U2	Using <application name> would improve my job performance.	✔	✔			✔
U3	Using <application name> in my job would increase my productivity.	✔	✔	✔	✔	✔
U4	Using <application name> would enhance my effectiveness on the job.	✔	✔		✔	✔
U5	Using <application name> would make it easier to do my job.	✔	✔	✔	✔	✔
U6	I would find <application name> useful in my job.	✔	✔	✔		✔
Perceived Ease of Use						
E1	Learning to operate <application name> would be easy for me.	✔	✔	✔	✔	N/A
E2	I would find it easy to get <application name> to do what I want it to do.	✔	✔			N/A
E3	My interaction with <application name> would be clear and understandable.	✔	✔			N/A
E4	I would find <application name> to be flexible to interact with.				✔	N/A
E5	It would be easy for me to become skillful at using <application name>.		✔	✔	✔	N/A
E6	I would find <application name> easy to use.	✔	✔	✔	✔	N/A

* In his second study, Adams (Study 2, 1992) added a new item so that the ease-of-use construct would have as many items (six) as in Davis's original model.

**Segars and Grover (1993) recommended a model with three latent factors (i.e., usefulness, effectiveness & ease of use). Items U2 and U4 were identified as measures of effectiveness rather than usefulness.

***Chin and Todd (1995) only focused on usefulness measures; their recommendations were not applicable (N/A) to the ease-of-use construct.

of concern whenever data-driven model modifications are carried out (MacCallum, Roznowski, & Necowitz, 1992), especially when the sample sizes are small for confirmatory factor analysis. Thus, despite several cross-validation efforts, tests of the validity and reliability of these instruments have not been definitive.

There is also considerable controversy concerning the robustness or measurement equivalence (invariance) of the perceived usefulness and ease-of-use instruments. Segars and Grover (1993) questioned the usefulness of these instruments for making comparisons. They cautioned that "no absolute measures for these constructs exist across varying technological and organizational contexts" (p. 525). Subramanian (1994) disagreed. In a study of two applications in two organizations, he found that the perceived usefulness and ease-of-use constructs were robust and concluded that IS researchers can use these instruments in varying technological and organizational contexts. However, Subramanian's sample sizes and application diversity were limited, prohibiting his findings from being definitive. Thus, we know little about the robustness (i.e., measurement equivalence) of these instruments, and this lack of knowledge limits the usefulness of these instruments for both research and practical decision making.

Invariance Hypotheses

If the perceived usefulness and ease-of-use instruments are to be used to make comparisons across applications and user population groups, the instruments should be invariant across type of application or technology, the user's general experience level with computing, and gender. There are a priori reasons to suspect that variations in application type, general experience with computing, and gender may influence the user's frame of reference and, thereby, the meaning or scaling of perceived usefulness or ease-of-use items.

Software packages are tools that are used for the purposes/tasks of the end-user. The concept of "usefulness" implies a purpose. Applications can be considered "useful" if they contribute to accomplishing the end-user's purpose, and "easy to use" if the effort required to make use of the application is modest relative to the end-user's frame of reference. Goodhue (1992) referred to purpose as the end-user's task. He argued that task characteristics (through system-task fit) influence user evaluations.

Four types of applications are in wide use in nearly every personal productivity software package: spreadsheets, word processing, database, and graphics. These applications are used for different purposes (i.e., objectives): data analysis, writing, inquiry/reporting, and presentation preparation, respectively. They also differ in the effort required to use the application. The purpose of the application or its comparative advantage (i.e., relative effort over alternative means of meeting the user's task requirements) may affect the user's frame of reference for responding to "usefulness" or "ease-of-use" items. "Usefulness" or "ease of use" may not have the same meaning for a spreadsheet user as it does for a user preparing a graphic presentation. For example, in graphic applications, the ability to select color schemes and change them easily may be implicit criteria for judging usefulness or ease of use. A spreadsheet user's criteria might focus more on formula application and manipulation than color selection. Previous cross-validation studies

have been careful to analyze responses separately for each type of application, implicitly suggesting that there may be difficulty comparing scores. This suggests the following hypotheses:

H1-U: The perceived usefulness instrument is invariant across word processing, graphic, database, and spreadsheet applications.

H1-E: The perceived ease-of-use instrument is invariant across word processing, graphic, database, and spreadsheet applications.

End users vary widely in their general experience with computing. Users with no computing experience may have difficulty forming an opinion of an application's usefulness or ease of use. They have no experience with other applications that can provide a point of comparison. After brief exposure, their opinion or attitude towards a specific application may be incomplete (i.e., in the formative stage).

Experienced users can be expected to have a frame of reference that is based upon more extensive computing experience. They are likely to have tried a greater number of, and perhaps better, software packages as well as explored a greater range of functionality within the same software packages. Thus, they might have higher standards than novices or inexperienced users for evaluating usefulness or ease of use.

Several researchers have suggested that a user's experience level may have an impact on their frame of reference. Adams et al. (1992) discussed how user perceptions of the ease-of-use and usefulness constructs might vary with a user's level of experience with computing. Igbaria et al. (1995) found that the level of computing experience had a significant direct effect on perceptions of usefulness and ease of use. The research on computer interfaces suggests that novices prefer direct manipulation interfaces (icons) whereas experienced users prefer command line interfaces (Bailey, 1982; Dix, Finlay, Abowd, & Beale, 1993). This suggests that the user's general experience with computing may affect their frame of reference for evaluating "usefulness" or "ease of use."

H2-U: The perceived usefulness instrument is invariant across users with no prior computing experience, novices, and experienced users.

H2-E: The perceived ease-of-use instrument is invariant across users with no prior computing experience, novices, and experienced users.

In the social sciences, gender has been widely studied as a key demographic factor that affects one's orientation or frame of reference towards phenomena (Eagly, 1987). Eagly contended that the division of labor between the sexes shapes gender roles in such a way that women are expected to behave relatively communally (e.g., showing concern for the welfare of other people) and men relatively agentically (e.g., showing competency, task orientation, instrumentality). As both males and females respond to these social expectations and act out their roles, they develop gender-based skills and beliefs (i.e., frames of reference). Thus, sex roles may be a determinant of a respondent's frame of reference for evaluating usefulness or ease of use, resulting in nonequivalent scales that cannot be compared.

Self-efficacy is one of the beliefs that comprise an individual's frame of reference (Bandura, 1977, 1982). Self-efficacy is a self-appraisal of one's ability to perform a task successfully. Computer self-efficacy is one's self-appraisal of his or her ability to perform a computer task. Miura (1984) reported that males report significantly higher computer self-efficacy (i.e., higher self-reported computer competency) than females. Computer self-efficacy and perceived ease of use are similar concepts (Davis, 1989), but they have different units of analysis (individual versus application). Gender differences in computer self-efficacy might be expected to affect an individual's perceptions of an application's usefulness and, especially, its ease of use. This suggests the following hypotheses:

H3-U: The perceived usefulness instrument is invariant across gender.

H3-E: The perceived ease-of-use instrument is invariant across gender.

The MIS literature provides considerable motivation for testing these hypotheses. Several studies have reported that females have significantly less positive attitudes/opinions towards computing than males (Dambrot, Watkins-Malek, Silling, Marshall, & Garver, 1985; Gilroy & Desai, 1986; Harrison & Rainer, Jr., 1992; Levin & Gordon, 1989; Parasuraman & Igbaria, 1990; Wilder, Mackie, & Cooper, 1985). Typically, these differences between the sexes in attitudes or beliefs are statistically significant, but quite small in an absolute sense (Dambrot et al.; Wilder et al.). None of these studies have tested the instruments they used for measurement invariance. If gender affects frame of reference, some of the findings may have to be reinterpreted or reexamined (e.g., the findings may be an artifact of scale differences).

RESEARCH METHODS

The main question in assessing the functional equivalence of an instrument is: Does the set of measures relate to a set of underlying constructs in the same way and to the same extent in multiple groups (Cole & Maxwell, 1985)? There are three levels of functional equivalence. At the first level, the issue is whether the instrument taps the same constructs in each subgroup. This first level is called *conceptual* or *congeneric equivalence* and is indicated by testing for *equal factor pattern* across subgroups (Bollen, 1989). No restrictions are placed upon the metrics or reliabilities of measures across subgroups. The principle concern is whether the items truly measure the same factors in disparate subgroups. Substantial between-group differences may have important practical or theoretical implications for researchers or managers who use these instruments to predict usage.

The second issue in assessing functional equivalence is whether the measures relate to a set of underlying constructs to the same extent in multiple subgroups (i.e., equivalent item-factor loadings or true scores). Measures with invariant true scores (but different errors) are termed *robust* or *tau-equivalent* (Bollen, 1989). Tau-equivalency is assumed whenever effect sizes are compared across groups as in MANOVA. The equality of item-factor loadings is the minimum condition for factorial invariance. If this condition is not met, the constructs have different meanings across groups and tests of other measurement parameters

may be meaningless. Substantial differences in item-factor loadings between groups indicate that the perceived usefulness and ease-of-use constructs have to be scaled differently for problematic groups, and comparisons between groups are difficult, if not impossible.

A less important issue in functional equivalency is the comparability of reliabilities across groups. Observed scores that have equivalent errors (as well as equivalent true scores) are called *fully equivalent*. Full equivalency across subgroups is not required by MANOVA, but greatly disparate reliabilities between groups would suggest that the utility of an instrument to account for true variance of the construct is restricted in one or more subgroups.

Sample

In order to gauge accurately the invariance of the perceived usefulness and ease-of-use instruments, a wide range of subjects representing a variety of backgrounds, demographics, education, and experience were surveyed during a lab experiment to test the technology acceptance model. The sample comprised students in introductory undergraduate courses in information systems at two large Western and Midwestern universities. The courses were part of the core curriculum required of all business students and included a cross section of majors including marketing, management, finance, economics, hotel administration, human resources/industrial management, accounting, and management information systems. The student populations enrolled at each institution include a significant portion of nontraditional students (approximately 25%) with work experience.

The demographic dimensions included type of application (spreadsheet, database, word processing or graphics), prior experience with computers in general (no experience, novice or experienced), and gender (male, female). The number of observations by demographic dimension varies due to respondent omissions. Of the 902 responses, 244 were spreadsheet, 156 were database, 292 were word processing, and 210 were graphics. For the 581 responses to the prior general computing experience question, 105 categorized themselves as having "no experience," 244 as "novices," and 232 as "experienced." For gender, 355 were females and 371 were males.

Experimental Treatment

Students were assigned to an application with which they had no prior application-specific experience. Then, they were grouped by assigned application and given a brief (one hour) demonstration of the software. During the demonstration, students were exposed to the breadth of functionality possible via the application's use. Subjects were not allowed to use the software during this demonstration. The researchers wanted to make sure that all subjects were focusing on the functionality being presented rather than "playing" with the software. Students were instructed that the software application could be utilized to complete requirements in subsequent courses, as well as in their future employment environments.

Immediately following this demonstration, subjects were given another brief (one-half hour) period to experience their assigned application for themselves in a hands-on manner. During this time, subjects were allowed to do any type of

experimentation they wished. No assignment was given because: (1) the application types were different, and (2) the researchers did not want the subjects to feel that they "failed" to complete any assigned task. Upon completion of this hands-on session, the perceived usefulness/ease-of-use questionnaires were administered. Thus, subjects were responding based upon their informed perceptions.

The data represent "initial exposure" responses in which the user did not have previous application-specific experience with the software package. Thus, the lab experiment simulated the early development or prepurchase trial context for which the technology acceptance model was designed (Davis et al., 1989). Favorable psychometric properties under conditions of relatively brief initial exposure would be encouraging relative to their use as early warning indicators of user acceptance (Ginzberg, 1981).

The researchers feel that findings based on student data are transferable to the referent population of business respondents involved in early development or prepurchase decisions for office suite applications. Goodhue (1992) argued that system, task, individual characteristics, organizational characteristics, and beliefs about the expected consequences of use influenced user evaluations of systems. Student findings are transferable because they use the same office suite applications as business respondents, and they use them for the same purposes: to store, retrieve and analyze data, prepare presentations, and write papers. The office suite applications are generic tasks. Thus, system and task characteristics would be similar. Organizational characteristics should be less important with these generic office tasks.

The two respondent groups overlap and merge over time. They overlap because many of the students in the sample (i.e., approximately 25%) are nontraditional and have at least some business experience. They merge over time because today's business students become tomorrow's business respondents. This means that individual characteristics of business respondents may lag the business student population, but the two groups should have many individual characteristics in common. Today's students may be inexperienced. This inexperience may limit generalizability, especially in tests involving nongeneric applications (e.g., specialized software used in a particular business). Where the application is generic (e.g., office suite applications), we would expect student beliefs about the expected consequences of use to be comparable to business respondents who are also given a brief prepurchase exposure to a similar software package used for the same generic task. To help ensure this, students were told that the software applications could be used to complete requirements in subsequent courses, as well as in their future employment environments.

The use of business students in a lab setting enabled researchers to control the initial exposure, insuring uniform treatment. It also enabled researchers to control the types of applications studied. There is precedent for using student responses to validate these instrument(s). To develop these instruments, Davis (1989) used students in a lab setting for the same reason (i.e., to get responses based on brief initial exposure to the target systems). Finally, given the large sample sizes required for invariance analysis, the use of student subjects was considered a practical necessity (i.e., problems of access, timing, and treatment control would make it difficult to get large numbers of initial exposure respondents from business organizations).

Measurement Model

Invariance analysis requires a prior specification of a measurement model. Due to the small sample sizes used in the previous cross-validation studies, their conflicting recommendations, and the problem of capitalization on chance, the researchers felt that the original measurement model proposed by Davis had not been, as yet, adequately assessed. Therefore, Davis's original measurement model (all 12 items in the diagram depicted in Figure 1) was used in this research. The item format and scales were the same as those used by Davis (see Table 2).

Confirmatory Analysis Methods

Confirmatory factor analysis was conducted using LISREL VIII (Jöreskog & Sörbom, 1993). Davis's hypothesized measurement model (see Figure 1) was tested for fit against the sample of 902 observations. LISREL is a statistical tool for analyzing covariance matrices according to systems of structural equations. Model specification is accomplished by fixing or constraining elements in three matrices that are analogous to the factor pattern matrix, factor correlation matrix, and communalities from a common factor analysis.

Because no one statistic is universally accepted as an index of model adequacy, our interpretation of results emphasizes substantive issues, practical considerations, and several measures of fit. Although the chi-square statistic is a global test of a model's ability to reproduce the sample variance/covariance matrix, it is sensitive to sample size and departures from multivariate normality (Bollen, 1989). Thus, the chi-square statistic must be interpreted with caution (Jöreskog & Sörbom, 1993). Even if the discrepancy between the estimated model and the data is very small, if the sample size is large enough, almost any model will be rejected because the discrepancy is not statistically equal to zero.

Fit indexes have been developed to assess the degree of congruence between the model and data (i.e., whether the variance and covariance in the data are accounted for by the model). Wheaton (1987) has suggested that it is prudent to report several fit measures. This research reports CFI (Bentler, 1990) and NNFI (Bentler & Bonnet, 1980).

CFI is a normed relative noncentrality index. It estimates each noncentrality parameter by the difference between its T statistic and the corresponding degrees of freedom. The Tucker-Lewis index (Tucker & Lewis, 1973) was among the earliest fit indexes that involved comparing a model's fit relative to other nested models. Tucker and Lewis' original purpose for developing their index was to quantify the degree to which a particular exploratory factor model is an improvement over a zero-factor model when assessed by maximum likelihood. Bentler and Bonnet's (1980) nonnormed fit index (NNFI) is a generalization of Tucker and Lewis' definition to all types of covariance-structured models under various estimation methods.

The CFI and NNFI indexes are used because they are generally unaffected by sample size. Medsker, Williams, and Holahan (1994) recommended the CFI index as being the best approximation of the population value for a single model. Marsh, Balla, and McDonald (1988) reported that the NNFI is also generally unaffected by sample size; it is useful for situations in which a parsimony-type index is needed to account for the number of estimated parameters in a model (Medsker et

al.). Good-fitting models generally yield fit indexes of at least .90, that is, only a relatively small amount of variance remains unexplained by the model (Bentler & Bonnet, 1980; Bollen, 1989).

In confirmatory factor analysis, factor loadings can be viewed as regression coefficients in the regression of observed variables on latent variables. Item-factor loadings are estimates of the validity of the observed variables (items). The larger the factor loadings or coefficients, as compared with their standard errors and expressed by the corresponding t-values, the stronger is the evidence that the measured variables represent the underlying constructs (Bollen, 1989). Completely standardized factor loadings above .70 contain more trait variance than error variance. Thus, in this research, items with loading above .70 will be considered to have good construct validity. Confirmatory factor analysis also enables us to estimate the reliability of individual items. The proportion of variance (R-square) in the observed variables that is accounted for by the latent variables influencing them can be used to estimate the reliability of the observed variables (items). Items with R-square values above .50 will be considered to have good reliability.

Invariance Analysis Methods

Confirmatory factor analysis (CFA) models of factorial invariance (Marsh, 1987, 1994) enable one to test explicitly the structure of a measurement model or its individual parameters for equivalence across subgroups or conditions. When parallel data exist for more than one group, CFA provides a particularly powerful test of the equivalence of solutions across the multiple groups. The researcher is able to fit the data subject to the constraint that any one parameter, any set of parameters, or all parameters are equal in the multiple groups.

Tests of factorial invariance across multiple groups involve a hierarchical ordering of nested models. Any two models are nested as long as the set of parameters estimated in the more restrictive model is a subset of the parameters estimated in the less restrictive model. Bentler (1990) and Bentler and Bonnet (1980) noted the usefulness of testing a series of nested models M_0, ...M_i, ...M_k, ..., M_s in which M_0 is a suitably defined null model, M_s is the saturated model with $df = 0$, and M_i, and M_k are models with positive df of intermediate complexity. When one model is a subset of another larger model, the difference between two models can be tested by subtracting the two chi-square values and testing this value against the critical value associated with the difference in degrees of freedom (Bentler & Bonnet; Long, 1983).

This chi-square test is very powerful and, where the hypothesis of equal factor loadings or error terms is not rejected, it provides strong support that observed differences between subgroups in parameters are due to chance (Marsh, 1987). If the chi-square difference is significant, the subjective fit indexes are examined to see how much they decline as invariance constraints are imposed. Small decreases in the subjective fit indexes would suggest that the differences in factor loadings or error terms are not substantial and are unlikely to effect the interpretation of research results.

To evaluate whether model-data fit declines substantially as invariance parameters are imposed, the researchers will examine changes in CFI, NNFI, and

the target coefficient. The target coefficient (Marsh, 1987) is a measure of the ability of a model with invariance constraints to explain the covariation compared to the corresponding model with no invariance constraints. It is defined as $(N - I)/(N - U)$, where N, I, and U are the chi-squares for the null model (N), the model with invariance constraints (I) and the corresponding model without any invariance constraints (U). Thus, the target coefficient provides an estimate of the proportion of the variance explained by the no invariance model that can be explained by the constrained (nested) model (Marsh & Hocevar, 1985).

Sequence of Invariance Analysis

The hierarchical sequence begins with the least restrictive model in which only the form of the model—the pattern of fixed and nonfixed parameters—is invariant across groups. This initial baseline model is "totally noninvariant" in the sense that there are no between-group invariance constraints on estimated parameters (i.e., a "no invariance" model). The examination of goodness-of-fit and parameter estimates within subgroups is essential to assessing the issue of congeneric or conceptual equivalence and identifying where, if anywhere, the model does not achieve "good fit."

This baseline model provides a basis of comparison for all subsequent models in the invariance hierarchy. If it is not able to fit the data, then none of the more restrictive models in the hierarchy will be able to do so (Marsh, 1994). The failure of the baseline model for a particular subgroup to achieve "good fit," or at least adequate fit, on subjective fit indexes may preclude any further tests of invariance involving this subgroup (Bollen, 1989). Poor fit in a subgroup suggests that the instrument may not measure the phenomena adequately in this subgroup, and a new instrument or measurement model may have to be developed.

Next, the equivalence of factor loadings across groups in each dimension was tested (i.e., tau-equivalency). In evaluating measurement models, the primary concern is usually whether each item is a good measure of its latent construct. Factor loadings are examined first because the equivalence of factor loadings is the minimal condition for "factorial invariance." Bollen (1989) noted that the equality of factor loadings is generally of a higher priority than the equality of other parameters. Bentler (1988) suggested testing first for invariance of factor loadings because, without such invariance, it would be difficult to argue that the factors are the same. If the factors are not the same, it may be meaningless to test for the invariance of other parameters. To test for equal factor loadings, an equal item-factor loading constraint was added to the baseline model, creating a nested or more restrictive model, which is a subset of the baseline model. Thus, the chi-square differences between these nested models provide a test of the hypotheses of equal item-factor loadings.

Finally, the researchers tested for the full equivalency (equal factor loadings and error terms) of the measurement model across the groups in each dimension. This nested or more restrictive model is also a subset of the baseline model. Thus, the chi-square differences between these nested models provide tests of full equivalency.

RESULTS

To conduct a confirmatory factor analysis, the researchers examined model-data fit and parameter estimates for the entire sample ($n = 902$). In the entire sample, the hypothesized measurement model has a chi-square of 599.69 for 53 degrees of freedom ($p = .0000$). Subjective fit indices indicate good fit: .94 and .95 for NNFI and CFI, respectively. The results indicate that the measurement model is appropriately specified, a proper solution is obtained, and the solution fits the entire sample well.

Parameter estimates and completely standardized parameter estimates for the overall sample of 902 respondents are illustrated in Table 3. The completely standardized item-factor loadings are high (above .70) and significant, as evidenced by their t-values. All items had reliability scores (R-square) above .50, indicating good reliability. Thus, all six of Davis's original perceived usefulness items and all six of his ease-of-use items are valid and reliable measures of their theoretical construct. Of all the items, Item E4 had the lowest factor loading and R-square scores, but these scores were still above the .70 and .50 criterions for good validity and reliability, respectively.

The researchers conducted a subgroups analysis of model-data fit. For each of the nine subgroups, Table 4 reports chi-square statistics, p-values, and subjective goodness-of-fit indices. A proper solution was obtained for each subgroup. Although each group has a significant chi-square, seven of the nine subgroups (spreadsheet, database, graphics, novices, experienced users, females, and males) have CFI and NNFI scores above .90, indicating "good" model-data fit. According to the subjective fit indexes, the word-processing subgroup does not achieve good model-data fit, but its CFI (.88) and NNFI (.86) scores could be considered "adequate" fit. According to the subjective fit indexes, the no-experience group has poor model-data fit (CFI = .79, NNFI = .74). These model-data fit results suggest that researchers should be cautious when using the perceived usefulness and ease-of-use instruments for word processing users, and especially cautious if the sample contains users with no prior computing experience.

Even for the seven groups that have good model-data fit, these results indicate that the same pattern of parameter estimates is able to fit the data from each group, but not that these parameter estimates take on the same, or even similar, values for the different groups. To test the instrument for tau-equivalence (equal factor loadings or true scores) and full equivalence (equal factor loadings and error terms) across type of application, level of experience, and gender, the researchers conducted a multigroup invariance analysis.

Because the word-processing and the no-experience subgroups did not have good model-data fit, the researchers first conducted invariance analysis with these subgroups included and then repeated the analysis with them excluded. Thus, the invariance analysis by type of application was run for both four groups (including word processing) and three groups (omitting the word-processing group). The experience dimension was analyzed in a similar manner: invariance analysis was conducted for both the three-group model (no experience, novice, experienced) and a two-group model, omitting the no-experience group.

Table 3: Parameter estimates and *t*-values for perceived usefulness and ease-of-use instrument ($n = 902$).

Items	Factor Loading	*t* value	Completely Standardized Factor Loading	Uniqueness/ Error Term	*R*-square (Reliability)
U1	1.00*		.87	.29	.76
U2	1.04	36.76	.88	.29	.77
U3	1.10	39.41	.91	.24	.83
U4	1.08	40.27	.91	.20	.83
U5	1.09	36.26	.87	.34	.76
U6	1.10	38.03	.89	.28	.79
E1	1.00*		.86	.29	.74
E2	1.08	36.68	.89	.25	.79
E3	1.00	37.95	.90	.18	.81
E4	.92	27.21	.75	.54	.56
E5	.95	32.60	.83	.32	.69
E6	1.04	37.13	.89	.22	.79

*Indicates a parameter fixed at 1.0 in the original solution.

The invariance analysis by type of application, level of experience, and gender is contained in Table 5a, 5b, and 5c, respectively. The invariance analysis (see Table 5) reports the chi-square and degrees of freedom for the equal pattern model (Model 1). The equal pattern model's chi-square and degrees of freedom are a baseline or point of comparison for the subsequent chi-square tests of nested models. For each nested model that imposes invariance constraints (i.e., Models 2-U, 2-E, 3-U, 3-E, and 4), Table 5 reports the change in chi-square from Model 1, the change in degrees in freedom from Model 1, and the corresponding *p*-value for the test of equivalent parameters across groups (e.g., tau-equivalency and full equivalency). Nonsignificant *p*-values mean the hypothesis of equal parameters is not rejected. Table 5 also reports three subjective fit indexes (NNFI, CFI, and the target coefficient). Small decreases in these subjective fit indexes suggest that the differences in factor loadings or error terms between subgroups are not substantial.

Invariance Analysis by Type of Application

Table 5a reports the results for the invariance analysis by type of application. The statistical significance of the difference in chi-square for two nested models can be evaluated relative to the difference in the degrees of freedom for the two models. Using four groups, the invariance analysis by type of application (Model 2-U) indicates that constraining the factor loading for the six items in the perceived usefulness instrument to be equal across subgroups resulted in a change in chi-square of 32 for 15 degrees of freedom (significant at $p = .0064$), rejecting the hypothesis that the usefulness factor loadings are equivalent across spreadsheet, database, word processing, and graphic applications. Holding the item-factor loadings

Table 4: Subgroup analysis of goodness of fit of perceived usefulness and ease-of-use instruments.

Dimensions	Subgroups	Chi-square	*df*	*p*-values	NNFI	CFI
Type of Application	Spreadsheet (n = 244)	268.73	53	.0000	.91	.93
	Database (n = 156)	128.27	53	.0000	.95	.96
	Word processing (n = 292)	408.74	53	.0000	.86	.88
	Graphics (n = 210)	162.27	53	.0000	.93	.95
Prior Experience	No experience (n = 105)	267.89	53	.0000	.74	.79
	Novice (n = 244)	222.14	53	.0000	.93	.94
	Experienced (n = 232)	192.71	53	.0000	.95	.96
Gender	Female (n = 355)	355.98	53	.0000	.90	.92
	Male (n = 371)	279.84	53	.0000	.95	.96

invariant for the six perceived ease-of-use items (Model 2-E) yielded a change in chi-square of 18 for 15 degrees of freedom; thus, the hypothesis of invariant factor loadings was not rejected (p = .2626). For four subgroups, these results suggest that the ease-of-use instrument is tau-equivalent across applications and the usefulness instrument is not.

For the perceived usefulness instrument, the NNFI, CFI, and target coefficient scores for Model 2-U can be compared to Model 1 to provide an indication of how substantial the changes in model-data fit are from holding the factor loadings invariant. The NNFI and CFI scores do not change between Model 1 and Model 2-U. The target coefficient score indicates that Model 2-U is .9966 as effective as Model 1 in explaining the covariance in the sample data. Thus, the difference between subgroups in factor loadings for the perceived usefulness instrument is statistically significant, but it does not appear to be substantial.

Full equivalency was tested by holding both factor loadings and error terms invariant for both the perceived usefulness (Model 3-U) and ease-of-use (Model 3-E) instruments. For 33 degrees of freedom, Models 3-U and 3-E have changes in chi-square of 293 (p = .0000) and 215 (p = .0000), respectively. Thus, the hypothesis of full equivalency is rejected for both instruments.

Model 4 assesses the change-in-fit indexes and the change in chi-square from the baseline model when all parameters of the measurement model in Figure 1 are held invariant across applications. The change in chi-square (637) for 75 degrees of freedom is significant (p = .0000). The NNFI and CFI indexes changed from .91 and .92 in Model 1, to .88 and .87 in Model 4, respectively. Thus, using an equivalent measurement model rather than an equal pattern model deteriorates fit below that required for good fit. The target coefficient for Model 4 indicates that the fully constrained model is .9320 as effective as Model 1 at explaining the covariance in the sample data.

This invariance analysis by type of application was redone with the word processing responses excluded (i.e., three groups) to see if the word processing group's "adequate" as opposed to "good" model-data fit explained the significant differences in factor loadings found in the four-group analysis. The results (see Table 5a,

Table 5: Invariance analysis of perceived usefulness and perceived ease-of-use instruments.

a. Invariance Analysis by Type of Application

Model Number	Model Description	Chi-square	*df*	Change in Chi-square from Model 1	Change in *df* from Model 1	*p*-value	NNFI	CFI	Target Coefficient
Type of Application (Four groups: Spreadsheet, Database, Word Processing, and Graphics)									
1	Equal Pattern	968	212	0	0		.91	.92	1.0000
2-U	Only Perceived Usefulness Factor Loadings Invariant	1000	227	32	15	.0064	.91	.92	.9966
2-E	Only Perceived Ease-of-use Factor Loadings Invariant	986	227	18	15	.2626	.91	.92	.9980
3-U	Only Perceived Usefulness Factor Loadings and Error Terms Invariant	1261	245	293	33	.0000	.89	.90	.9687
3-E	Only Perceived Ease-of-use Factor Loadings and Error Terms Invariant	1183	245	215	33	.0000	.90	.91	.9770
4	All Parameters Invariant for Usefulness and Ease of use	1605	287	637	75	.0000	.88	.87	.9320
Type of Application (Three groups: Spreadsheet, Database, and Graphics)									
1	Equal Pattern	559	159	0	0		.93	.94	1.0000
2-U	Only Perceived Usefulness Factor Loadings Invariant	574	169	15	10	.1320	.93	.94	.9977
2-E	Only Perceived Ease-of-use Factor Loadings Invariant	571	169	12	10	.2850	.93	.94	.9982
3-U	Only Perceived Usefulness Factor Loadings and Error Terms Invariant	756	181	197	22	.0000	.91	.92	.9702
3-E	Only Perceived Ease-of-use Factor Loadings and Error Terms Invariant	742	181	182	22	.0000	.91	.92	.9724
4	All Parameters Invariant for Usefulness and Ease of use	1056	209	497	50	.0000	.89	.88	.9249

Table 5: (continued) Invariance analysis of perceived usefulness and perceived ease-of-use instruments.

b. Invariance Analysis by Level of Experience

Model Number	Model Description	Chi-square	*df*	Change in Chi-square from Model 1	Change in *df* from Model 1	*p*-value	NNFI	CFI	Target Coefficient
Prior Experience (Three Groups: No Experience, Novice, and Experienced)									
1	Equal Pattern	683	159	0	0		.91	.93	1.0000
2-U	Only Perceived Usefulness Factor Loadings Invariant	717	169	34	10	.0002	.91	.92	.9949
2-E	Only Perceived Ease-of-use Factor Loadings Invariant	694	169	11	10	.3575	.92	.93	.9983
3-U	Only Perceived Usefulness Factor Loadings and Error Terms Invariant	849	181	166	22	.0000	.90	.91	.9754
3-E	Only Perceived Ease-of-use Factor Loadings and Error Terms Invariant	715	181	32	22	.0774	.92	.93	.9953
4	All Parameters Invariant for Usefulness and Ease of use	928	209	245	50	.0000	.91	.90	.9638
Prior Experience (Two Groups: Novice and Experienced)									
1	Equal Pattern	415	106	0	0		.94	.95	1.0000
2-U	Only Perceived Usefulness Factor Loadings Invariant	421	111	7	5	.2203	.94	.95	.9989
2-E	Only Perceived Ease-of-use Factor Loadings Invariant	419	111	5	5	.4156	.94	.95	.9992
3-U	Only Perceived Usefulness Factor Loadings and Error Terms Invariant	436	117	22	11	.0244	.94	.95	.9964
3-E	Only Perceived Ease-of-use Factor Loadings and Error Terms Invariant	427	117	12	11	.3636	.94	.95	.9980
4	All Parameters Invariant for Usefulness and Ease of use	478	131	63	25	.0000	.94	.94	.9894

Table 5: (continued) Invariance analysis of perceived usefulness and perceived ease-of-use instruments.

c. Invariance Analysis by Gender

Model Number	Model Description	Chi-square	*df*	Change in Chi-square from Model 1	Change in *df* from Model 1	*p*-value	NNFI	CFI	Target Coefficient
Gender (Two Groups: Female and Male)									
1	Equal Pattern	636	106	0	0		.93	.94	1.0000
2-U	Only Perceived Usefulness Factor Loadings Invariant	645	111	9	5	.1089	.93	.94	.9989
2-E	Only Perceived Ease-of-use Factor Loadings Invariant	650	111	14	5	.0156	.93	.94	.9984
3-U	Only Perceived Usefulness Factor Loadings and Error Terms Invariant	664	117	28	11	.0032	.93	.94	.9967
3-E	Only Perceived Ease-of-use Factor Loadings and Error Terms Invariant	684	117	49	11	.0000	.93	.94	.9943
4	All Parameters Invariant for Usefulness and Ease of use	773	131	137	25	.0000	.93	.93	.9839

Three-groups analysis by type of application, Model 2-U) indicate that with the word processing subgroup excluded, the hypothesis of equal factor loading for the perceived usefulness instrument is not rejected ($p = .1320$).

Invariance Analysis by Prior General Experience with Computing

The results for the invariance analysis by prior experience with computing are reported in Table 5b. The results indicate that the ease-of-use instrument is tau-equivalent and fully equivalent (at $p = .05$) across the three groups: no experience, novice, and experienced, but the usefulness instrument is only tau-equivalent across novice and experienced users. In the three-group model, NNFI does not decline as the equal factor loading constraint is imposed on the perceived usefulness instrument. CFI only changes from .93 to .92, and the target coefficient for Model 2-U is high (.9949), suggesting that the differences in factor loadings are statistically significant but, perhaps, depending upon the purpose of the researcher or decision maker, not substantial.

The exclusion of the no-experience group improves the goodness-of-fit scores for Model 4 as evidenced by NNFI, CFI, and target coefficient scores changing from .91, .90, and .9638 to .94, .94, and .9894, respectively. These results and the poor model-data fit for the no-experience subgroup suggest that researchers should be cautious in surveying users with no prior computing experience. The usefulness scores for users with no prior computing experience may be in a different scale and, thus, not directly comparable.

Invariance Analysis by Gender

The results of the invariance analysis by gender are reported in Table 5c. The tau-equivalency hypothesis was rejected for the ease-of-use instrument ($p = .0156$) but not for the usefulness instrument ($p = .1089$). The hypothesis of full equivalency was rejected for both ease of use ($p = .0000$) and usefulness ($p = .0032$). Although these results suggest that gender significantly affects the frame of reference by which ease of use is evaluated, there is evidence that this significance is due to the large sample sizes of the subgroups.

Measurement differences by gender are not substantial. The NNFI score remains at .93 for Model 1 through Model 4 and the CFI score remains at .94 for all models except Model 4 (CFI = .93). The target coefficient remains above .99 for Model 3-U and 3-E and falls no lower than .9839 for the fully invariant model (Model 4).

IMPLICATIONS FOR RESEARCH AND MANAGERIAL DECISION MAKING

This research has had two objectives. First, using a sample of "initial exposure" respondents, the researchers attempted to address the conflicting and equivocal findings reported in prior cross-validation studies. In the overall sample of 902 respondents, Davis's original measurement model achieved good model-data fit as evidenced by subjective fit indexes. Furthermore, all six items of the usefulness scale and all six items of the ease-of-use scale had good construct validity

and reliability. This is in marked contrast to the prior studies that eliminated items but disagreed on which items should be dropped (Adams et al., 1992; Segars & Grover, 1993; Subramanian, 1994).

These results have important implications for how the technology acceptance model is used in decision making. They show that individuals can develop well-formed beliefs about usefulness and ease of use after only a brief initial exposure. Thus, the technology acceptance model can be used as suggested in Davis et al. (1989) for prepurchase decisions or early evaluation decisions during the systems development process in which users have no application-specific experience. Both experienced and novice computer users can have well-formed beliefs about usefulness and ease of use. Thus, management need not rely only on experienced or expert users. People who have only a limited general experience with computers and consider themselves novices are able to evaluate a focal application after a brief initial exposure.

A second objective was to assess the invariance of Davis's original instruments across three dimensions: type of application, level of experience with computing in general, and gender. Table 6 (see columns 1& 2) summarizes the tests of the six hypotheses and reports findings. The implications for decision making are summarized in Column 3 of Table 6. The group names contained in a single box indicate the range of subgroups across which managers can be confident in making comparisons between scores. Subgroups that are not in the same scale (not comparable) are indicated by presenting the subgroup name in a separate box.

Invariance of the Ease-of-Use Instrument

This research has provided evidence that the ease-of-use instrument is tau-equivalent across type of application and level of experience. The hypothesis of tau-equivalency across gender was rejected, suggesting statistically significant but not necessarily substantial differences in true scores. Both the male and female subgroups had good model-data fit. From a practical decision-making perspective, the data suggests that the scale differences are inconsequential. For most practical purposes, managers can make comparisons between male and female respondents. This is indicated in Table 6 by separating group names (i.e., separate male and female boxes). The boxes are made contiguous to indicate that, for most practical decision-making purposes, the scores are comparable.

Female respondents have slightly lower item-factor loadings or true scores than males. This difference is small, but consistent across all six ease-of-use items. All six items have acceptable loadings for both genders. This suggests that ease of use has a similar meaning in both genders, but the scales are slightly different. For the same level of the latent variable (ease of use), females report slightly lower scores than males.

Decision makers should not place undue importance on females reporting slightly lower ease-of-use scores than males. If the difference is not large enough to be of practical importance in decision making, it may be only an artifact of measurement. The implication is that females should be trained and supported the same ways as males unless, for a particular application, one finds a substantial enough difference in ease-of-use scores to merit differential treatment.

This finding of nonequivalence of the ease-of-use instrument across gender is more problematic in research in which statistical significance is often considered an important finding. Using large samples, many of the studies on sex roles in computing report statistically significant, but not necessarily substantial, differences between men and women in computer-related attitudes/opinions (e.g., difficulty in using technology, computer anxiety, computer self-efficacy, etc.). Evidence that one attitude/opinion scale (i.e., ease of use) is not equivalent across gender suggests the need for further research to reexamine other attitude/opinion scales to make sure that statistically significant findings related to sex roles are not an artifact of gender differences in the scales.

Invariance of the Perceived Usefulness Instrument

The evidence presented here suggests that the perceived usefulness instrument is tau-equivalent across spreadsheet, database, and graphic applications, between female and male respondents, and among both novice and experienced users (see column three of Table 6). Usefulness scores across these application, gender, and level of experience categories are in the same scale. Thus, comparative statements across these subgroups can have substantive import and responses may be combined for data analysis. This finding greatly enhances the value of the usefulness instrument as a tool for decision making (i.e., benchmarking applications or user groups).

Individuals with no prior general computer experience can form opinions about an application's ease of use during a brief initial exposure, but they do not appear to develop well-formed perceptions of the application's usefulness. Such users appear to lack an experiential frame of reference or point of comparison for evaluating the software's usefulness.

Researchers and practitioners should be cautious in interpreting usefulness scores for word processing applications or pooling such responses with other types of applications. In initial exposure situations, perceived usefulness seems to have a different meaning for word processing respondents than for users of spreadsheets, graphics, or database applications. The reason for this difference is not clear. The usefulness items emphasize enhanced productivity rather than improved grammar (fewer spelling errors, fewer grammatical errors) or enhanced appearance (font selection, integrated text and graphics). In the context of word processing, usefulness may have a different meaning. It may mean appearance and/or grammatical accuracy as well as productivity. Future research may make a contribution by enhancing our theoretical understanding of why this nonequivalence occurs.

Despite these notable exemptions for word processing applications and users with no experience, the results show that, for most practical purposes, the perceived usefulness and ease-of-use instruments used in the technology acceptance model can be considered invariant across key office applications that are widely used. Knowledge that spreadsheet, database, and graphic applications are in the same scale and, therefore, directly comparable is an interesting and important finding that enhances the utility of these instruments. For example, it suggests that one can compare the usefulness or ease of use of a spreadsheet with a database package or a graphics application. This finding should: (1) enhance the value of these instruments by facilitating their use as tools for making early development or

Table 6: Summary of research results and managerial implications.

Hypothesis	Statistical Findings	Managerial Implications
Perceived Usefulness Results		
H1-U: The perceived usefulness instrument is invariant across word processing, graphics, database, and spreadsheet applications.	H1-U: Rejected. The word-processing subgroup had poor model-data fit. Thus, the hypothesis that perceived usefulness instrument is invariant across three applications (graphics, spreadsheet, and database) was tested and accepted.	Usefulness scores for graphics, spreadsheet, and database applications are comparable (i.e., in the same scale), but word process applications should be analyzed separately. Graphics, Spreadsheet, and Database | Word Processing
H2-U: The perceived usefulness instrument is invariant across users populations with no prior computing experience, novices, and experienced users.	H2-U: Rejected. The no prior-computing-experience subgroup had poor model-data fit. Thus, the hypothesis that perceived usefulness is invariant across two groups (novice and experienced users) was tested and accepted.	Usefulness scores for novice and experienced user populations are comparable (i.e., in the same scale), but responses from users with no prior computing experience are not. Novice, Experienced Users | No Experience
H3-U: The perceived usefulness instrument is invariant across gender.	H3-U: Accepted.	Usefulness scores for males and females are comparable (i.e. in the same scale). Female, Male

Table 6: (continued) Summary of research results and managerial implications.

Hypothesis	Statistical Findings	Managerial Implications
Perceived Ease-of-Use Results		
H1-E: The perceived ease-of-use instrument is invariant across word processing, graphics, database, and spreadsheet applications.	H1-E: Accepted.	Ease-of-use scores for graphics, spreadsheet, database, and word processing applications are comparable (i.e., in the same scale). Graphics, Spreadsheet, Database, and Word Processing
H2-E: The perceived ease-of-use instrument is invariant across users populations with no prior computing experience, novices, and experienced users.	H2-E: Accepted.	Ease-of-use scores are comparable (i.e., in the same scale) across users populations with no prior computing experience, novices, and experienced users. Novice, Experienced Users, and Users with no Prior Experience
H3-E: The perceived ease-of-use instrument is invariant across gender.	H3-E: Rejected. Both male and female subgroups had good model-data fit.	Ease-of-use scores are not comparable (i.e., in the same scale) across males and female subgroups. Since both subgroups had good model-data fit, the scale differences are probably inconsequential for most practical decision making purposes. Female | Male

prepurchase decisions, and (2) increase the additivity of research by enhancing confidence in the comparability of results across studies. [Received: April 10, 1997. Accepted: November 19, 1997].

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William J. Doll is a professor of MIS and strategic management at the University of Toledo. He holds a doctoral degree in business administration from Kent State University and has published extensively on information system and manufacturing issues in academic and professional journals including *Management Science, Communications of the ACM, MIS Quarterly, Academy of Management Journal, Decision Sciences,* and *Information Systems Research.* Dr. Doll has published on a variety of topics including computer-integrated manufacturing, executive steering committees, top management involvement in MIS development, strategic information systems, information systems downsizing, and end-user computing. He has developed instruments to measure a variety of constructs including user involvement, user documentation, and end-user computing satisfaction.

Anthony R. Hendrickson is an assistant professor of management at Iowa State University. He received his PhD in computer information systems and quantitative analysis from the University of Arkansas. Dr. Hendrickson's research interests include information systems usage, information systems psychometrics, and object-oriented technologies. He has published in a variety of journals including *MIS Quarterly, Decision Sciences, The Journal of Computer Information Systems, DATA BASE, Ethics and Critical Thinking Quarterly,* and *HR Magazine,* as well as other journals and conference proceedings. He is an active member of the Decision Sciences Institute, the Association for Computing Machinery, and a charter member of the Association of Information Systems.

Xiaodong Deng is a PhD student in manufacturing management in the College of Business at the University of Toledo. He holds an MS in systems analysis and a BS in computer science, both from Tsinghua University, Beijing, P.R. China. He was a lecturer and research assistant at Tsinghua University before he joined the University of Toledo. He has co-authored a book (*150 How-tos on the Usage of Personal Computers,* 1994, Guang-ming Daily Press), and co-translated a book (*IBM OS/2 2.0 Graphics Programming Reference Vol. II,* 1994, Tsinghua University Press). He has published an article on customer order processing and finished goods inventory in *New Techniques: Research and Applications,* a journal published by Tsinghua University.